

SMART DISPLAY MODULE SPECIFICATION

5.0 Inch Smart Display with TOUCH	
Model:	UEED80480050E-WB-A
Version:	V1.1
Date:	2025-03-12

Customer Confirmation

Approved by	Notes

REVISION HISTORY

[illegible]

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1. Introduction

1.1 Features

Brief Info:

- 1) Two buttons: a reset button and a boot button.
- 2) Backup IO: download ports and multiple IO leads to use on both sides of the periphery.
- 3) Power: DC 5V, 300mA

System

- 1) OS: RTOS
- 2) CPU: ESP32-S3 240Mhz
- 3) RAM: 8MB
- 4) Flash: 16MB
- 5) Interface: UART/USB
- 6) Support 2.4GHz Wi-Fi、BLE 5、BLE Mesh
- 7) Support Peripherals:
GPIO, SPI, LCD interface, Camera interface, UART, I2C, I2S, remote control, pulse counter, LED PWM, full-speed USB 2.0 OTG, USB Serial/JTAG controller, MCPWM, SDIO host, GDMA, TWAI® controller (compatible with ISO 11898-1), ADC, touch sensor, temperature sensor, timers and watchdogs

For more information on ESP32-S3-WROOM-1, please refer to the following link: [datasheet_en.pdf](#)

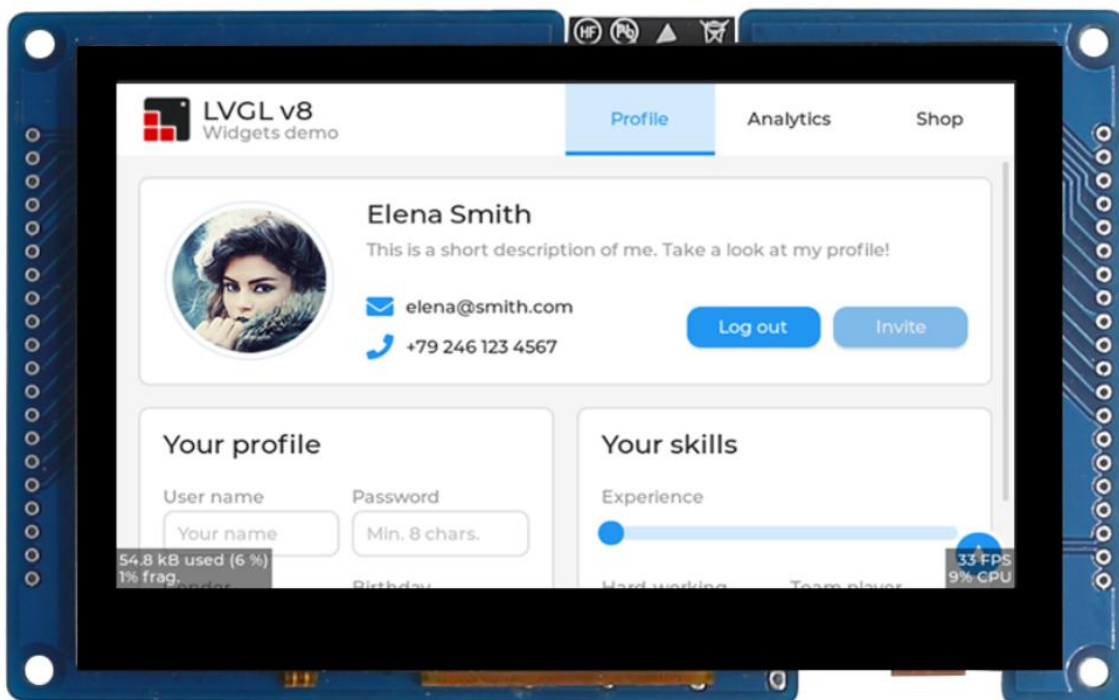
Display

- 1) Size: 5.0 Inch
- 2) Resolution: 800 *480
- 3) Pixel Driving element: Transflective TFT
- 4) Pixel Arrangement: RGB Vertical Stripe
- 5) Interface Mode: 40PIN RGB 24bits
- 6) Driver IC: JD9165 TP IC: GT911
- 7) Brightness: 200 cd/m²
- 8) Touch: CTP

Other

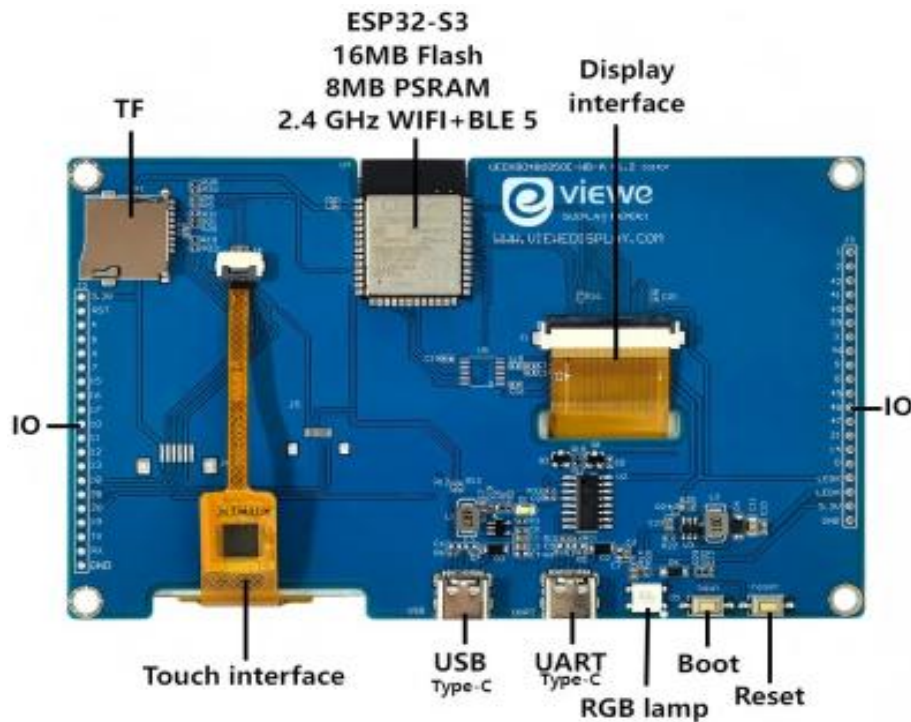
- 1) Operation Temperature: -20~70°C
- 2) Storage Temperature: -30~80°C

1.2 Appearance picture



2. Product information

2.1 Interface Description



Note: ① ②

- A pin can be used for other purposes when it is not used at the same time, When using Display interface, most of the pins are occupied because the RGB interface is used.
- You can also use an external gpio to drive other interface types, such as SPI interface, MCU interface, MIPI interface, etc., without using the Display interface provided by us
- Where GPIO0 is the link to the boot button, but you can use it for other purposes; GPIO17 and GPIO18 are not used, you are free to use them
- If the Display interface is used but the SD is not used, then the SD pins are freely usable.

① Display Interface:

Pin No.	Symbol	I/O	Description
1	LEDK	P	Power supply for backlight cathode
2	LEDA	P	Power supply for backlight anode
3	GND	P	Power Ground
4	VDD	P	Power supply to the internal logic power regulator(3.3V)
5-12	R0-R7	I	Red data input.
13-20	G0-G7	I	Green data input.
21-28	B0-B7	I	Blue data input.

29	GND	P	Power Ground
30	CLK	I	Pixel clock input pin, Negative polarity
31	DISP	I	Standby mode. Normally pulled high.
32	HSYNC	I	Horizontal sync signal, Negative polarity
33	VSYSN	I	Vertical sync signal, Negative polarity
34	DEN	I	Data input enable. Display access is enabled when DE is "H"
35	NC	I	Dummy
36	GND	P	Power Ground
37	XR	-	Dummy
38	YD	-	Dummy
39	XL	-	Dummy
40	YU	-	Dummy

I: Input; O: Output; P: Power

② TP Interface:

Pin No.	Symbol	I/O	Description
1	GPIO20	P	TP SCL
2	GPIO19	P	TP SDA
3	GPIO18	P	INT(It's not actually used)
4	GND	P	Power Ground
5	VDD	I	Power supply to the internal logic power regulator(3.3V)
6	GPIO38	I	RTP-csb-CTP-rst
7	GND	P	Power Ground
8	GND	P	Power Ground

2.2 Display Information

Item	Specification	Unit	Remark
Pixel Driving element	Transflective TFT	-	-
Screen Size	5.0	Inch	Diagonal
Resolution	800(W)*3(RGB)*480(H)	Dots	-
Module Power Consumption	1.719	Watt	With BL
Active Area	108(W)*64.8(H)	mm	-
Pixel pitch (W*H)	0.135(W)*0.135(H)	mm	-
Module Size (W*H*D)	120.8(W)*75.9(H)*2.73(D)	mm	-
Driver IC	JD9165	-	-
Interface Mode	40PIN RGB 24bits	-	-
Color gamut	40%@Transmissive, 16.5%@Reflective	%	Typ.
Display mode	Normally Black, Transflective	-	-
Luminance	Trans:200	cd/m ²	Typ.
Viewing Direction	U/D/L/R Trans: Typ. 60/60/60/50 (CR>2) Ref: Typ. 70/60/70/60 (CR>2)	Deg.	-
Display Color	16.7M	Colors	24bits
LED life time	25,000 Hours	-	-

2.3 Voltage & Current

Item	Conditions	Min	Typ	Max	Unit
Power Voltage	DC	4.0	5.0	5.5	V
Operation Current	VCC= +5V, Maximum backlight current		300		mA
	VCC= +5V,backlight off	-	150	-	mA
Recommended power supply:5V 1A DC					

2.4 Reliability Test

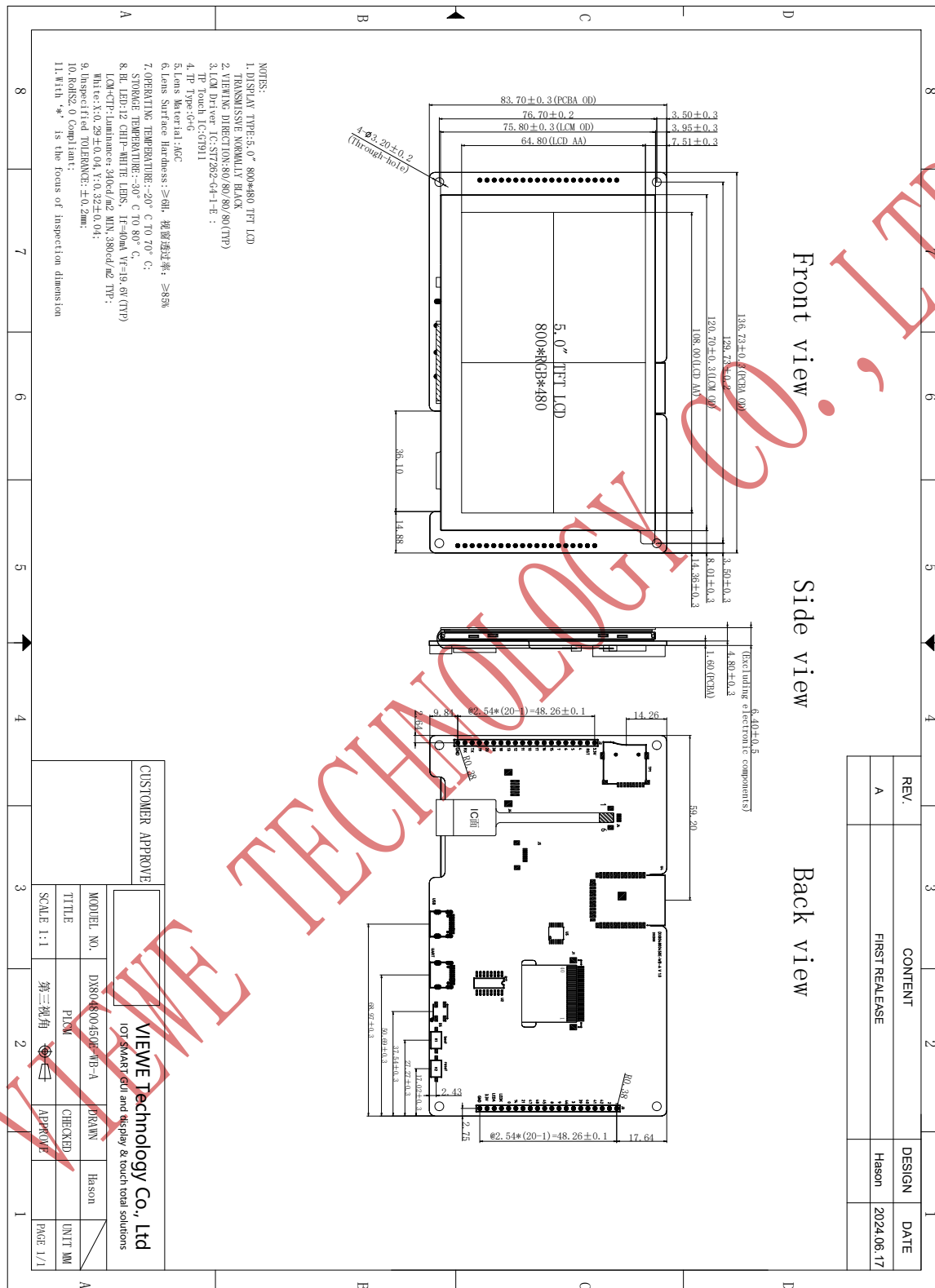
Item	Conditions	Min	Typ	Max	Unit
Working Temperature	60%RH at 5V voltage	-20	25	70	C
Storage Temperature	---	-30	25	85	C
Working Humidity	25°C	10%	60%	90%	RH
ESD	---	Contact: ±4KV Air: ±8KV			KV

2.5 Related software

Software name	Version	Software associated configuration
Arduino IDE	2.0.17 (esp32)	1. Board: ESP32S3 Dev Module 2. CPU Frequency: 240MHz (WiFi) 3. Flash Frequency: NO 4. Flash Mode: QIO 80MHz 5. Flash Size: 16MB (128Mb) 6. Partition Scheme: Default 4MB with spiffs (1.2MB APP/1.5MB SPIFFS) 7. PSRAM: OPI PSRAM 8. Programmer: Esptool
ESP-IDF	5.1.1 5.2.2	Once configured, no configuration is required (If you have any problem with the configuration, please contact us, we will help you)

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3. MECHANICAL DRAWING



4. Schematic



5. Related downloads

5.1 Arduino and IDF relevant information

<https://github.com/VIEWESMART/UEDX80480050ESP32-5inch-Touch-Display>

5.2 Libraries required for Arduino

<https://github.com/VIEWESMART/UEDX80480050ESP32-5inch-Touch-Display/tree/main/Libraries>

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